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(54) **ELECTRONIC DEVICE INCLUDING A CIRCUIT STORING A SIGNAL**

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See application file for complete search history.

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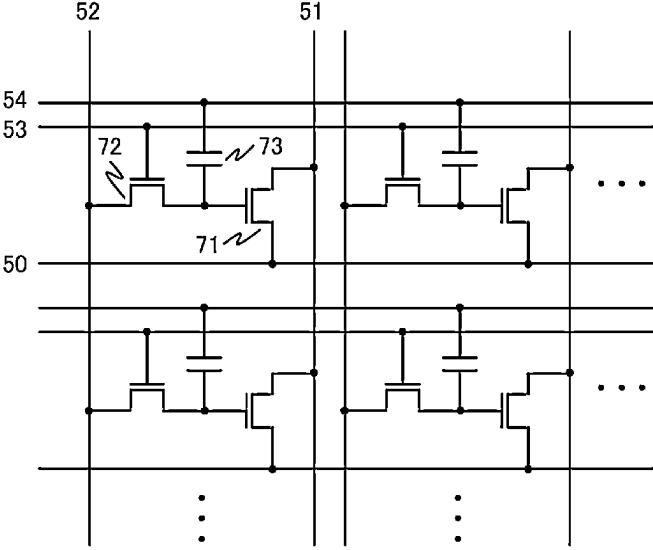
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(57) **ABSTRACT**  
A portable electronic device that can operate even when electric power supplied through contactless charge by electromagnetic induction is low is provided. The portable electronic device includes a reflective liquid crystal display which includes a transistor including an oxide semiconductor, a power source portion which includes a rechargeable battery capable of charge by contactless charge, and a signal processing portion which includes a nonvolatile semiconductor memory device. In the portable electronic device, electric power stored in the rechargeable battery is used in the reflective liquid crystal display and the signal processing portion.

**4 Claims, 10 Drawing Sheets**



| Related U.S. Application Data                                                                                                                                                                                                                                                                                                                                                                                 |  |  |              |      |                                          |
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FIG. 1

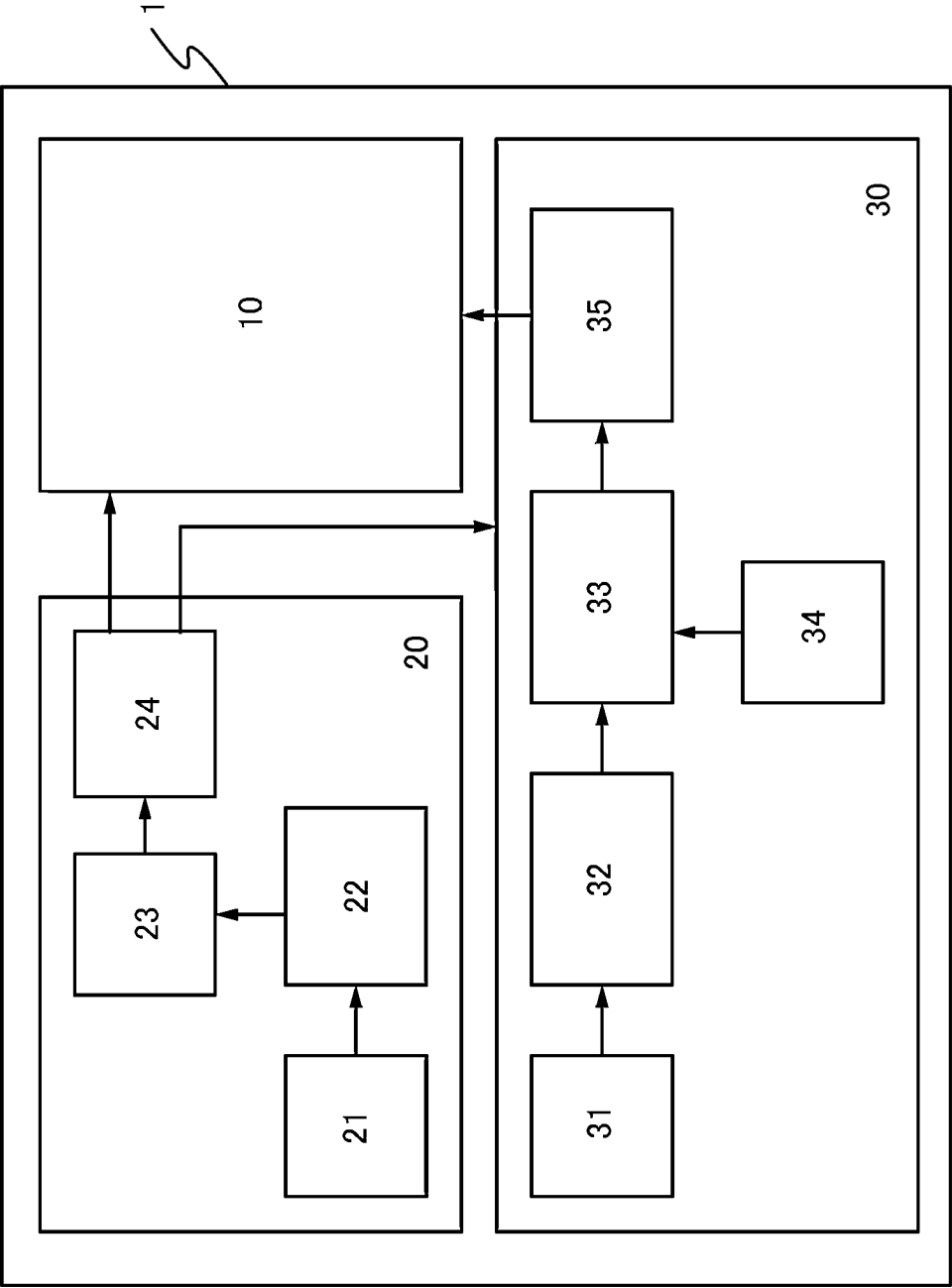


FIG. 2

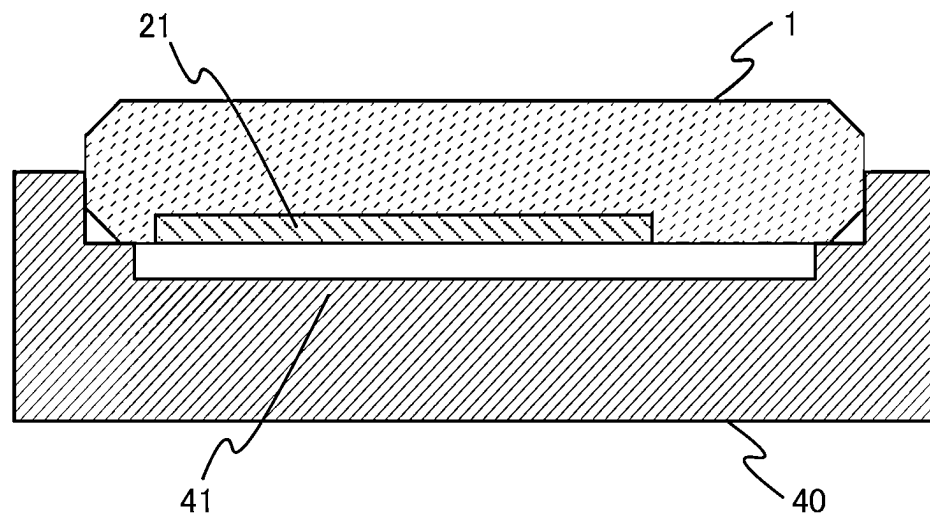


FIG. 3

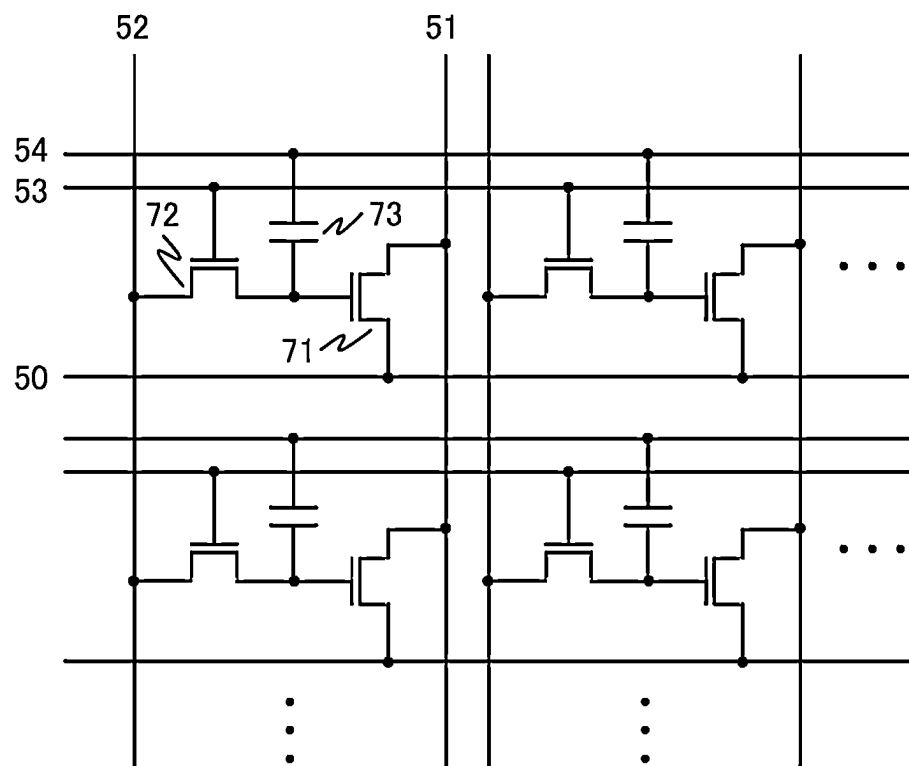




FIG. 4A

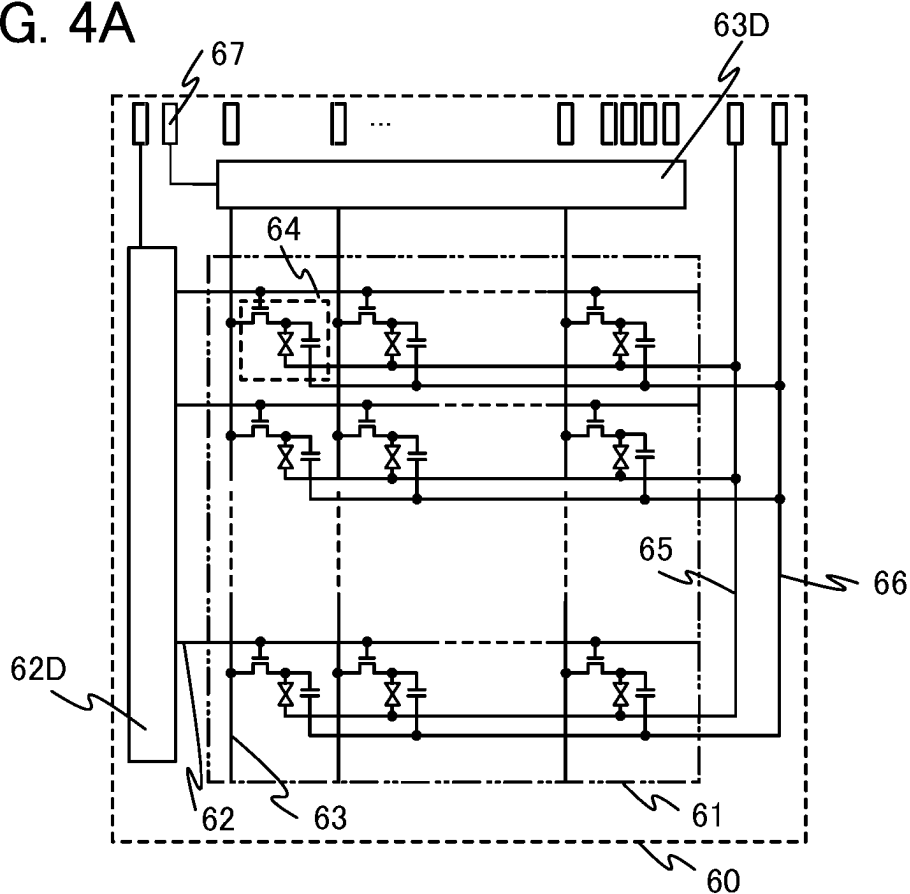


FIG. 4B

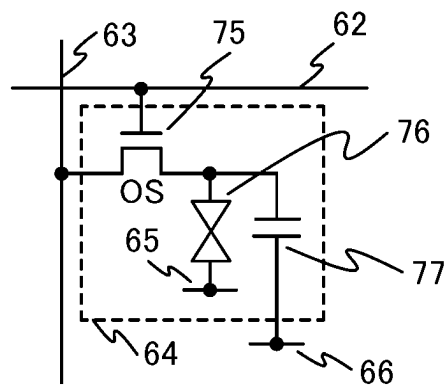


FIG. 4C

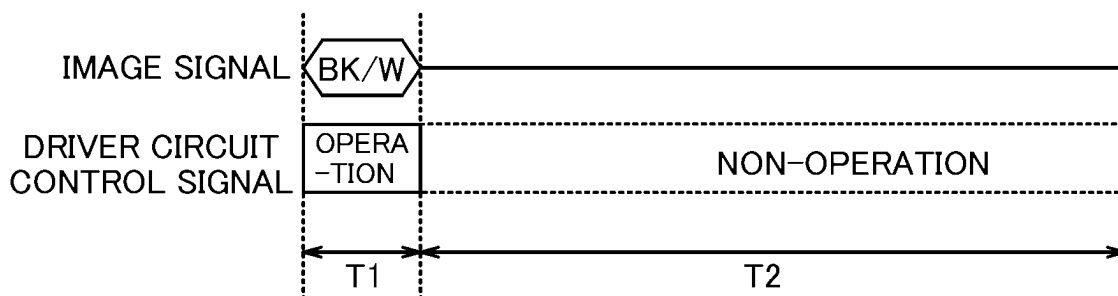


FIG. 5

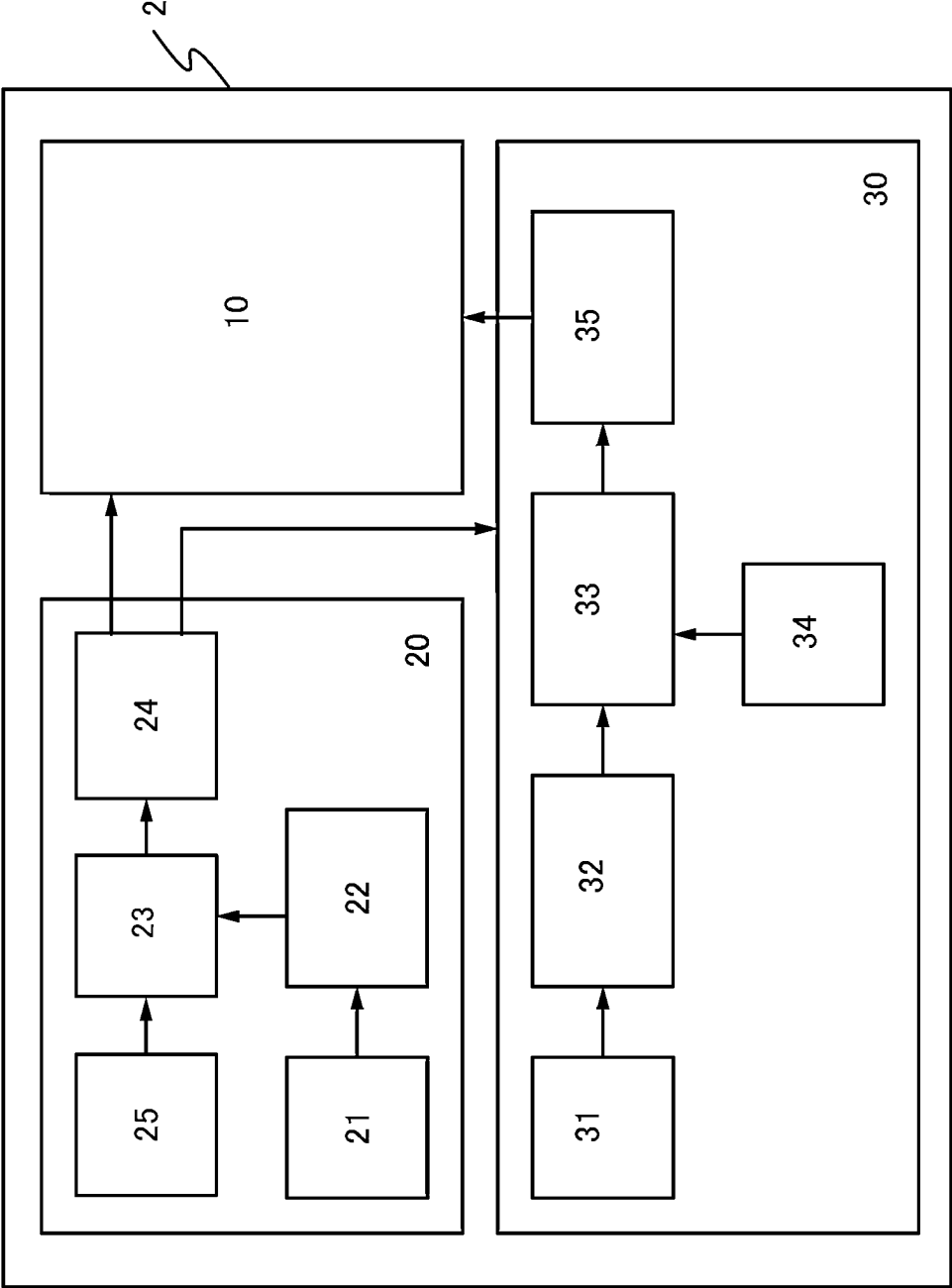


FIG. 6

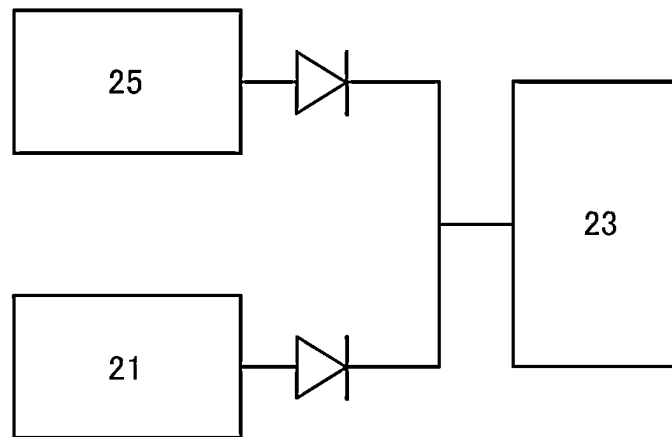


FIG. 7

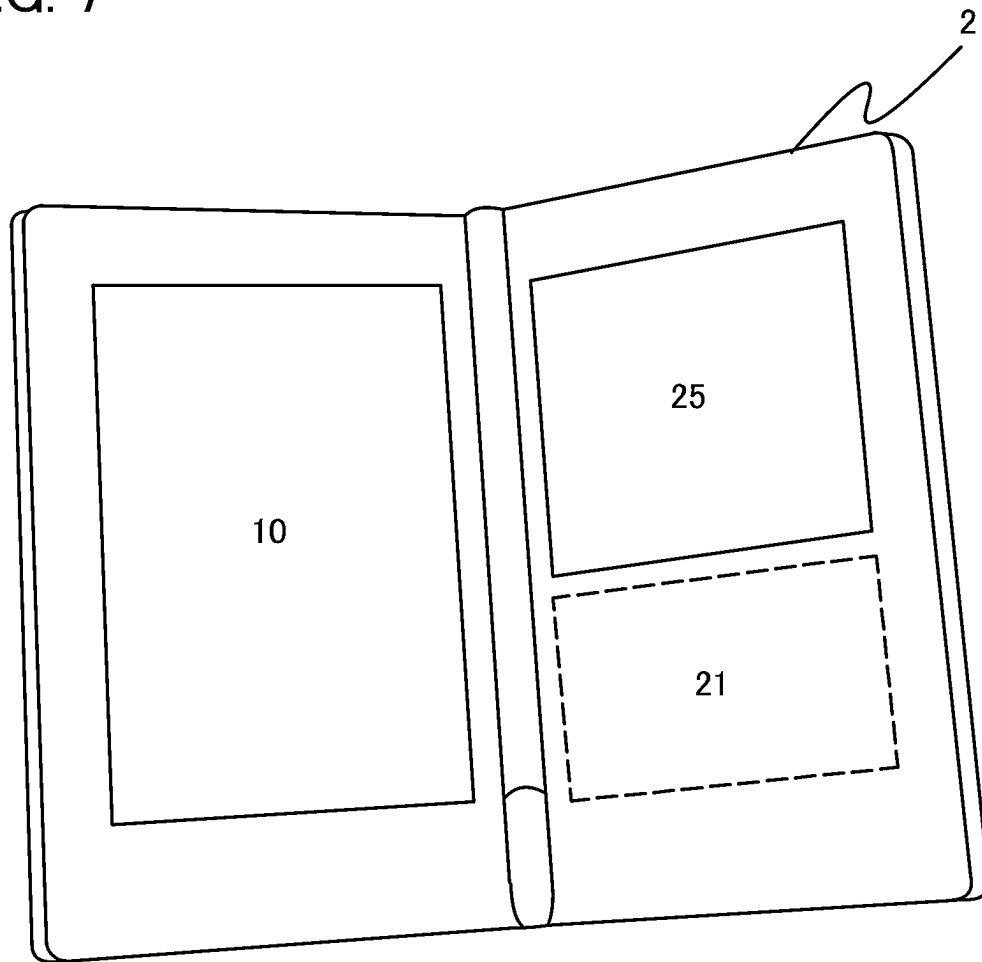


FIG. 8A

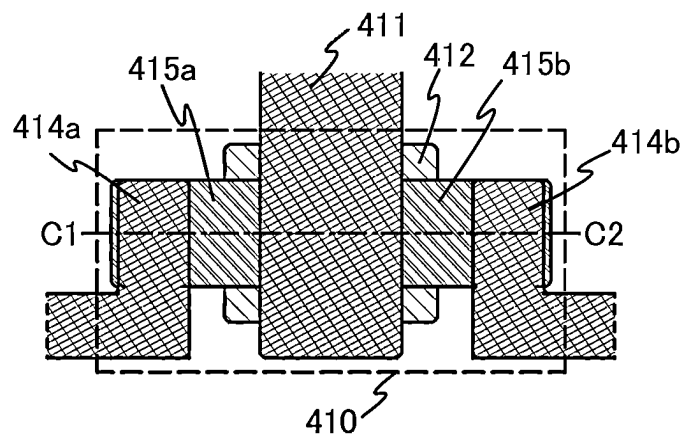


FIG. 8B

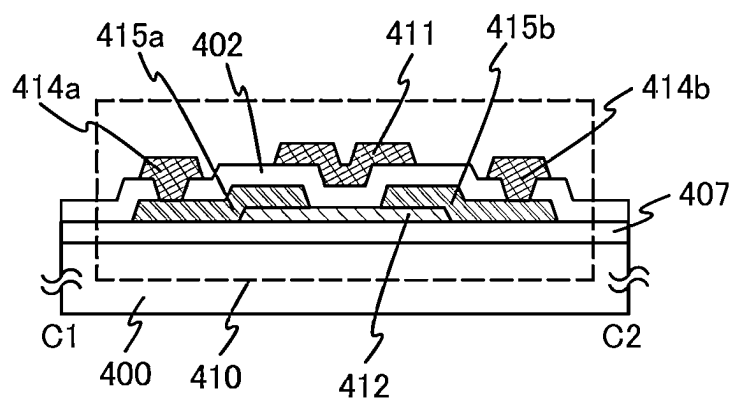


FIG. 9A

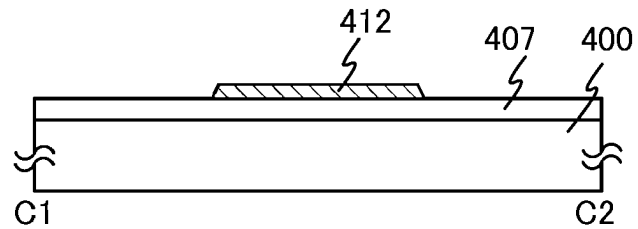


FIG. 9B

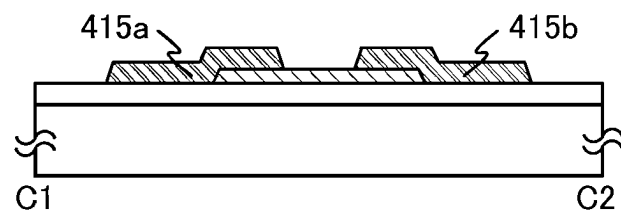


FIG. 9C

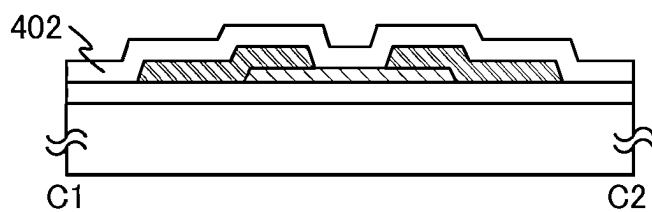


FIG. 9D

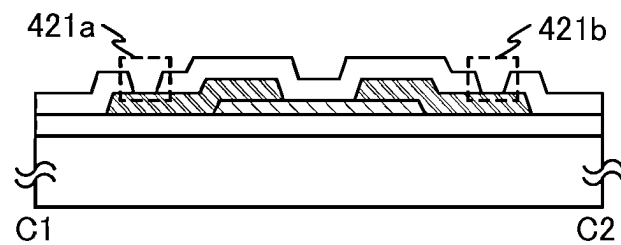


FIG. 9E

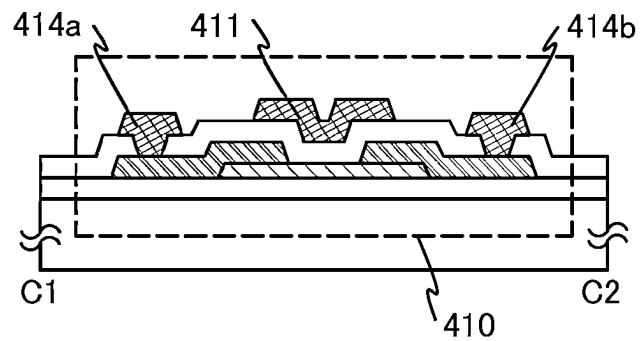
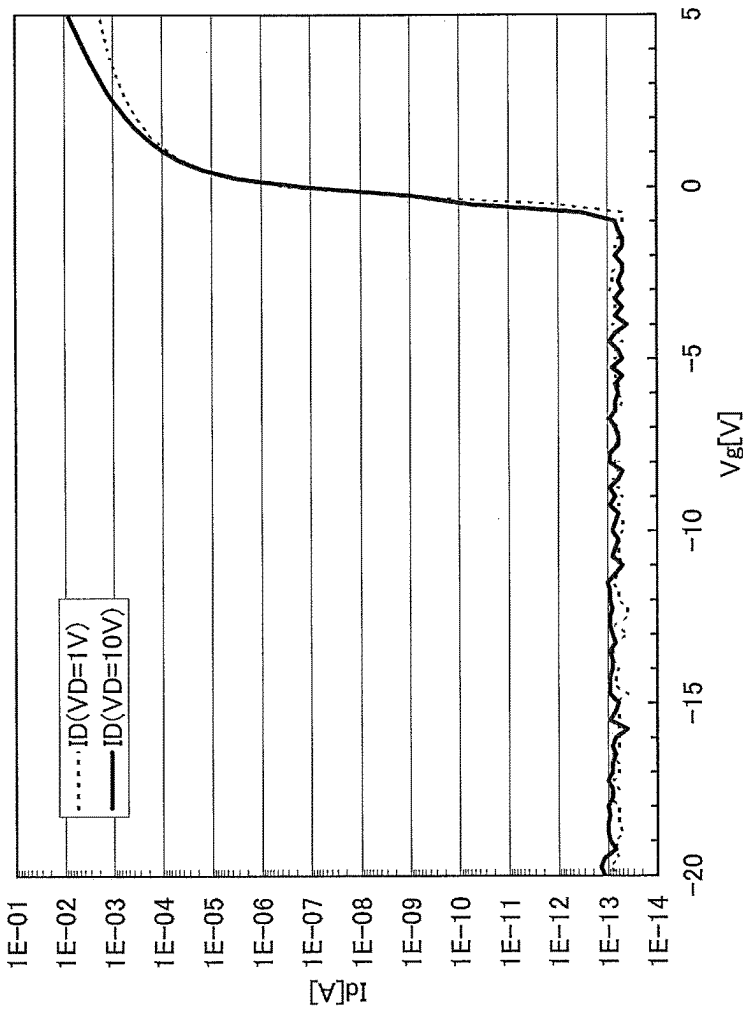


FIG. 10



1

# ELECTRONIC DEVICE INCLUDING A CIRCUIT STORING A SIGNAL

## BACKGROUND OF THE INVENTION

### 1. Field of the Invention

The technical field of the present invention relates to a portable electronic device capable of contactless charge.

### 2. Description of the Related Art

Portable electronic devices such as e-book readers that enable reading by displaying computerized book data on a display have been spread. Such devices are assumed to be used portably, and can be driven by batteries incorporated in main bodies. Therefore, these devices are preferably supplied with electric power by using an external power supply device.

In Patent Document 1, an e-book reader supplied with power supply voltage by using an AC adapter is suggested.

In Patent Document 2, a display device is suggested. The display device includes a display portion, a console portion for operation and input from external, an antenna portion for transmitting and receiving a wireless signal, a controller portion for controlling a signal that is input in the console portion and a signal that is transmitted and received by the antenna portion, and a battery portion for converting a wireless signal received by the antenna portion into electric power and for storing it as electric power for driving the display portion.

## REFERENCE

### Patent Document

[Patent Document 1] Japanese Published Patent Application No. 2007-147871

[Patent Document 2] Japanese Published Patent Application No. 2008-181108

## SUMMARY OF THE INVENTION

In order to supply power supply voltage from an AC adapter, a contact such as a metal electrode needs to be provided on a portable electronic device side. The contact needs durability. In addition, it is necessary to take measures against contact failure or generation of leakage due to short circuit, moisture, or the like.

In addition, in order that a portable electronic device can operate even when electric power supplied through contactless charge by electromagnetic induction is low, power consumption of the portable electronic device needs to be reduced.

An object of one embodiment of the present invention is to provide a portable electronic device that can operate even when electric power supplied through contactless charge by electromagnetic induction is low.

One embodiment of the present invention is a portable electronic device including a reflective liquid crystal display, a power source portion, and a signal processing portion. The reflective liquid crystal display includes a transistor including an oxide semiconductor, the power source portion includes a rechargeable battery capable of charge by contactless charge, and the signal processing portion includes a nonvolatile semiconductor memory device.

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Another embodiment of the present invention is a portable electronic device including a reflective liquid crystal display, a power source portion, and a nonvolatile semiconductor memory device. The reflective liquid crystal display includes a first transistor including a first oxide semiconductor, the power source portion includes a rechargeable battery capable of charge by contactless charge, and the nonvolatile semiconductor memory device includes a second transistor, a third transistor, and a capacitor. A first terminal of the second transistor is electrically connected to a first wiring, a second terminal of the second transistor is electrically connected to a second wiring, and a gate of the second transistor is electrically connected to a first terminal of the third transistor and one electrode of the capacitor. A second terminal of the third transistor is electrically connected to a third wiring, and a gate of the third transistor is electrically connected to a fourth wiring. The other electrode of the capacitor is electrically connected to a fifth wiring.

The third transistor may be a transistor including a second oxide semiconductor. The second oxide semiconductor may be the same as the first oxide semiconductor or different from the first oxide semiconductor.

Electric power stored in the rechargeable battery may be used in the reflective liquid crystal display and the signal processing portion.

The power source portion may include a solar cell.

The oxide semiconductor is intrinsic or substantially intrinsic, and the off-state current per unit channel width of a transistor is lower than or equal to 100 aA/ $\mu\text{m}$  ("a" is  $10^{-18}$ ), preferably lower than or equal to 1 aA/ $\mu\text{m}$ , more preferably lower than or equal to 1 zA/ $\mu\text{m}$  ("z" is  $10^{-21}$ ) at room temperature. Note that in this specification, "intrinsic" means the state of a semiconductor with a carrier concentration of lower than  $1 \times 10^{12}/\text{cm}^3$ , and "substantially intrinsic" means the state of a semiconductor with a carrier concentration of higher than or equal to  $1 \times 10^{12}/\text{cm}^3$  and lower than  $1 \times 10^{14}/\text{cm}^3$ .

Since contactless charge can be performed, a contact such as a metal terminal on a portable electronic device side does not need to be provided. In addition, a portable electronic device that can operate even when electric power supplied through contactless charge by electromagnetic induction is low can be provided.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of a portable electronic device.

FIG. 2 is a drawing of a structure example of the portable electronic device and a battery charger.

FIG. 3 is a diagram of an example of a circuit structure of a nonvolatile semiconductor memory device.

FIGS. 4A and 4B are schematic views of an example of a display panel of a liquid crystal display device, and FIG. 4C is a chart of an example of a method for driving the display panel.

FIG. 5 is a block diagram of a portable electronic device.

FIG. 6 is a diagram of a connection relation between an antenna, a solar cell, and a rechargeable battery.

FIG. 7 is a drawing of a structure example of the portable electronic device.

FIGS. 8A and 8B are drawings of a structure example of a transistor.

FIGS. 9A to 9E are drawings showing an example of a method for manufacturing the transistor.



FIG. 10 is a graph showing electric characteristics of transistors.

### DETAILED DESCRIPTION OF THE INVENTION

Embodiments of the invention disclosed herein will be described below with reference to the drawings. Note that the invention is not limited to the following description, and those skilled in the art can easily understand that modes and details of the invention can be changed in various ways without departing from the purpose and the scope of the invention. Accordingly, the invention should not be interpreted as being limited to the following description of the embodiments.

#### Embodiment 1

In this embodiment, a structure of hardware of a portable electronic device of this embodiment is described with reference to FIG. 1, FIG. 2, FIG. 3, and FIGS. 4A to 4C.

FIG. 1 is the block diagram of the portable electronic device of this embodiment. A portable electronic device 1 includes a display 10, a power source portion 20, and a signal processing portion 30. Note that an example in which the portable electronic device 1 is used as an e-book reader (also referred to as an e-book terminal device) is described below.

(Power Source Portion 20)

The power source portion 20 includes an antenna 21, a rectifier circuit 22, a rechargeable battery 23, and a DC-DC converter 24.

(Antenna 21)

The antenna 21 is a power-receiving coil for contactless charge.

In FIG. 2, a structure example of the portable electronic device and a battery charger are illustrated. As illustrated in FIG. 2, the portable electronic device 1 includes the antenna 21, and a battery charger 40 includes a power-sending coil 41. The contactless charge is performed by contactless transmission of electric power by electromagnetic induction between the coils, and a metal terminal is not provided between the coils.

The contactless charge does not need a contact such as a metal terminal for charging between the portable electronic device 1 and the battery charger 40. Therefore, there is no contact failure. Alternatively, leakage or the like due to short circuit, moisture, or the like hardly occurs.

(Rectifier Circuit 22)

Electric power received by the antenna 21 is rectified in the rectifier circuit 22 and is stored in the rechargeable battery 23.

(Rechargeable Battery 23)

The rechargeable battery 23 is a capacitor that can store electricity by charging and can be used repeatedly as a battery. As the rechargeable battery 23, for example, a lithium ion battery, a lithium ion capacitor, or the like can be used. In addition, the rechargeable battery 23 includes a control circuit for preventing overcharge and overdischarge.

(DC-DC Converter 24)

Electric power stored in the rechargeable battery 23 is transmitted to the display 10 and the signal processing portion 30 via the DC-DC converter 24 and is used as power supply voltage therein.

(Signal Processing Portion 30)

The signal processing portion 30 includes an antenna 31, a down converter 32, a signal processing circuit 33, an NVM (a nonvolatile memory) 34, and a display controller 35. (Antenna 31)

5 The antenna 31 transmits a request for starting download of computerized book data (referred to as "e-book data" below) which is selected by a user of the portable electronic device 1 to a server, and receives the e-book data sent from the server in response to the request.

10 (Down Converter 32 and Signal Processing Circuit 33)

The e-book data received by the antenna 31 is down-converted in the down converter 32 and converted into a baseband signal. The baseband signal is processed in the signal processing circuit 33.

15 (NVM 34)

Since the amount of the e-book data received by the antenna 31 is larger than that of data for one screen of the display, the received data is stored in the NVM 34.

20 Here, an example of the structure of a circuit of a nonvolatile semiconductor memory device that can be applied to the NVM 34 is described with reference to FIG. 3.

A nonvolatile semiconductor memory device illustrated in FIG. 3 includes a transistor 71, a transistor 72, and a capacitor 73.

25 A transistor including a material other than an oxide semiconductor is suitable for the transistor 71. Examples of the material other than the oxide semiconductor include single crystal silicon, crystalline silicon, and the like. The transistor including the material other than the oxide semiconductor can operate at high speed, so that data can be read from the nonvolatile semiconductor memory device at high speed.

30 In contrast, a transistor including an oxide semiconductor is suitable for the transistor 72. A transistor including an oxide semiconductor has a characteristic of extremely low off-state current. Thus, data stored in the nonvolatile semiconductor memory device can be retained for a long time. Therefore, refresh operation is unnecessary or performed less often, and power consumption of the nonvolatile semiconductor memory device can be reduced.

35 A gate of the transistor 71 is connected to one of a source and a drain of the transistor 72, a source of the transistor 71 is connected to a wiring 50 (a source line), and a drain of the transistor 71 is connected to a wiring 51 (a bit line). The other of the source and the drain of the transistor 72 is connected to a wiring 52 (a first signal line), and a gate of the transistor 72 is connected to a wiring 53 (a second signal line). In addition, one electrode of the capacitor 73 is connected to the gate of the transistor 71 and the one of the source and the drain of the transistor 72, and the other electrode of the capacitor 73 is connected to a wiring 54 (a word line).

40 Next, data write operation to the nonvolatile semiconductor memory device illustrated in FIG. 3 is described.

First, voltage is applied to the gate of the transistor 72, which is connected to the wiring 53, so that the transistor 72 is turned on. As a result, the wiring 52 and the transistor 71 are brought into conduction, and voltage is applied to the gate of the transistor 71 and the capacitor 73, so that data is written.

45 After that, the transistor 72 is turned off, so that the voltage applied to the gate of the transistor 71 is held and the data is stored. At this time, when a voltage at which the transistor 71 is turned on is applied to the gate of the transistor 71, the on state of the transistor 71 is held for a long time. In contrast, when a voltage at which the transistor

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71 is turned off is applied to the gate of the transistor 71, the off state of the transistor 71 is held for a long time.

Next, data read operation from the nonvolatile semiconductor memory device illustrated in FIG. 3 is described.

In the case of the on or off state of the transistor 71, a constant voltage is applied to the wiring 50 and a read voltage is applied to the wiring 54. As a result, in the case where the transistor 71 is on, the voltage of the wiring 51 varies. In contrast, in the case where the transistor 71 is off, the voltage of the wiring 51 does not vary. Therefore, by comparing the voltages between the wiring 50 and the wiring 51, the data stored in the nonvolatile semiconductor memory device can be read.

Data rewrite operation to the nonvolatile semiconductor memory device illustrated in FIG. 3 is similar to the data write operation.

(Display Controller 35)

The display controller 35 transmits data whose content is to be displayed on the display 10.

Most of the e-book data is still-image data; thus, the data rewrite speed does not need to be very high. Accordingly, high-speed operation of the down converter 32, the signal processing circuit 33, the NVM 34, and the display controller 35 is not required. Therefore, these circuits can be driven with a low voltage. Thus, even when electric power supplied through contactless charge by electromagnetic induction is low, the portable electric device 1 can operate.

(Display 10)

The display 10 displays content of the e-book data. Note that with the use of a display that does not emit light by itself, such as a reflective liquid crystal display or an electrophoretic display, power consumption of the display 10 can be set to 10 mW or less. Thus, even when electric power supplied through contactless charge by electromagnetic induction is low, the portable electric device 1 can operate.

Here, an example of a circuit structure that can be applied to a display is described with reference to FIGS. 4A and 4B.

FIG. 4A is a schematic view of an example of a display panel of a liquid crystal display device. A display panel 60 includes a pixel portion 61, a gate signal line 62, a gate signal line driver circuit 62D, a data signal line 63, a data signal line driver circuit 63D, a pixel 64, a common electrode 65, a capacitor line 66, and a terminal portion 67.

FIG. 4B is a view of the pixel 64 illustrated in FIG. 4A. The pixel 64 includes a transistor 75 including an oxide semiconductor, a liquid crystal element 76, and a storage capacitor 77.

Next, an example of a method for driving the display panel is described with reference to FIG. 4C.

First, in order to write an image signal BK/W to the pixel, the transistor 75 is turned on, and a period T (hereinafter referred to as a "writing period T1") during which a voltage based on the image signal is applied to a pixel electrode of the liquid crystal element 76 is provided. In the writing period T1, a driver circuit control signal is supplied to a driver circuit of the display 10 and the display controller 35, so that these circuits operate.

After the writing period T1, voltage  $V_{pix}$  is generated in the pixel electrode of the liquid crystal element 76. Then, the transistor 75 is turned off, so that the voltage  $V_{pix}$  is held in the pixel electrode of the liquid crystal element 76.

In a subsequent period T2 (hereinafter referred to as a "holding period T2") during which the voltage  $V_{pix}$  is held in the pixel electrode of the liquid crystal element 76, the image signal BK/W is not written. In addition, the driver

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circuit control signal is not supplied to the driver circuit of the display 10 and the display controller 35, so that these circuits do not operate.

The length of the holding period T2 varies depending on an off-state current  $I_{75}$  of the transistor 75 and a current  $I_{76}$  flowing through the liquid crystal element 76. In order to prevent a screen flicker due to variation in these currents, refresh operation by which data displayed on the screen is rewritten at regular intervals is necessary.

Note that the off-state current  $I_{75}$  of the transistor 75 including an oxide semiconductor is extremely low. Therefore, the length of the holding period T2 is dependent only on the current  $I_{76}$  flowing in the liquid crystal element 76. Thus, the number of rewrite operation of data to be displayed on the screen can be approximately  $1/1000$  the number of general rewrite operation (60 times per second).

As described above, in the holding period T2, the operation of the driver circuit of the display 10 and the display controller 35 can be stopped. Thus, power consumption of the display 10 and the display controller 35 can be approximately  $1/1000$  general power consumption.

Next, the oxide semiconductor included in the transistor is described.

Impurities such as hydrogen, moisture, hydroxyl group, or hydroxide (also referred to as a hydrogen compound) which serve as donors are intentionally removed from the oxide semiconductor included in the transistor, and then oxygen which is simultaneously reduced in the step of removing these impurities is supplied, so that the oxide semiconductor is highly purified and becomes electrically i-type (intrinsic). This is for suppressing variation in electric characteristics of the transistor.

Hydrogen contained in the oxide semiconductor is removed as much as possible; thus, the carrier density of the oxide semiconductor is less than  $1 \times 10^{14}/\text{cm}^3$ , preferably less than  $1 \times 10^{12}/\text{cm}^3$ , or more preferably less than  $1 \times 10^{10}/\text{cm}^3$ .

In an oxide semiconductor, which is a wide band gap semiconductor, the density of the minority carrier is low and the minority carrier is difficult to be induced. Thus, in the transistor including the oxide semiconductor, tunnel current is difficult to be generated; consequently, off-state current is difficult to flow.

In addition, impact ionization and avalanche breakdown are less likely to occur in the transistor including the oxide semiconductor, which is a wide band gap semiconductor. Therefore, the transistor including the oxide semiconductor has resistance to hot carrier deterioration. The hot carrier deterioration is mainly caused by an increase in the number of carriers due to avalanche breakdown and by injection of the carriers accelerated to high speed to the gate insulating film.

Note that the "off-state current" in this specification means current flowing between a source and a drain of an n-channel transistor with positive threshold voltage ( $V_{th}$ ) when a given gate voltage of higher than or equal to  $-20$  V and lower than or equal to  $-5$  V is applied at room temperature. Note also that "room temperature" is higher than or equal to  $15^\circ\text{C}$ . and lower than or equal to  $25^\circ\text{C}$ .

The transistor including the oxide semiconductor disclosed in this specification has a current value per micrometer channel width ( $1\ \mu\text{m}$ ) of  $100\ \text{aA}/\mu\text{m}$  or less, preferably  $1\ \text{aA}/\mu\text{m}$  or less, more preferably  $10\ \text{zA}/\mu\text{m}$  or less at room temperature.

As described above, by using an oxide semiconductor that is highly purified and becomes electrically i-type (intrinsic), a transistor with an extremely small off-state current value can be provided. A test element group (also referred to as a

TEG) is manufactured and the measurement results of off-state current characteristics are described below.

In the TEG, two hundred transistors each with  $L/W=3$   $\mu\text{m}/50$   $\mu\text{m}$  (the thickness  $d$  was 30 nm) are connected in parallel to provide a transistor with  $L/W=3$   $\mu\text{m}/10000$   $\mu\text{m}$ . Note that  $W$  represents the channel width and  $L$  represents the channel length.

FIG. 10 is a graph showing electric characteristics ( $\log(I_d)-V_g$ ) of the transistors provided in the TEG. In FIG. 10, the horizontal axis represents gate voltage  $V_g$  [V], and the vertical axis represents drain current  $I_d$  [A]. Note that the substrate temperature was room temperature, and the voltage  $V_d$  between a source and a drain was either 1 V (a dotted line in the graph) or 10 V (a solid line in the graph). In the above condition, the voltage  $V_g$  between the source and the gate was varied from  $-20$  V to  $+20$  V, and transfer characteristics of the drain current  $I_d$  were measured.

As shown in FIG. 10, the off-state current value of the transistor with a channel width  $W$  of  $10000$   $\mu\text{m}$  was  $1 \times 10^{-13}$  A or less when  $V_d$  was 1 V or 10 V. This value is less than or equal to the resolution (100 fA) of a measurement device (a semiconductor parameter analyzer, Agilent 4156C manufactured by Agilent Technologies, Inc.). This off-state current value corresponds to 10 aA/ $\mu\text{m}$  per micrometer channel width (1  $\mu\text{m}$ ).

#### Embodiment 2

In this embodiment, a structure of hardware of a portable electronic device which is different from the structure of hardware of the portable electronic device of Embodiment 1 is described with reference to FIG. 5, FIG. 6, and FIG. 7.

FIG. 5 is a block diagram of the portable electronic device of this embodiment. A portable electronic device 2 has a structure in which a solar cell 25 is added to the power source portion 20 of the portable electronic device 1 illustrated in FIG. 1. With the structure including the solar cell 25, when the portable electronic device 2 is exposed to sunlight or illumination light, electricity can be stored in the rechargeable battery 23 with the use of the solar cell 25.

In FIG. 6, a connection relation between the antenna, the solar cell, and the rechargeable battery are illustrated. Back-flow prevention diodes are provided between the antenna 21 and the rechargeable battery 23 and between the solar cell 25 and the rechargeable battery 23.

A structure example of the portable electronic device including the solar cell is described with reference to FIG. 7.

The portable electronic device 2 illustrated in FIG. 7 has a structure in which e-book data is displayed on the display 10 in the state where the portable electronic device 2 is open. Thus, when the solar cell 25 is provided so as to be exposed outside in the state where the portable electronic device 2 is open, the solar cell 25 can perform charge while the e-book data is displayed on the display 10.

A display that does not emit light by itself, such as a reflective liquid crystal display or an electrophoretic display, is particularly suitable for the display 10 because the solar cell 25 can perform charge.

#### Embodiment 3

In this embodiment, a structure example of a transistor including an intrinsic or substantially intrinsic oxide semiconductor and an example of a manufacturing method thereof are described with reference to FIGS. 8A and 8B and FIGS. 9A to 9E.

FIGS. 8A and 8B are an example of a plan structure and an example of a cross-sectional structure of a transistor. FIG. 8A is a plan view of a top gate transistor. FIG. 8B is a cross-sectional view of a portion taken along straight line C1-C2 in FIG. 8A.

A transistor 410 includes, over a substrate 400, an insulating layer 407, an oxide semiconductor layer 412, a first electrode 415a, a second electrode 415b, a gate insulating layer 402, and a gate electrode 411. The first electrode 415a is in contact with a wiring layer 414a, and the second electrode 415b is in contact with a wiring layer 414b.

Note that the transistor 410 illustrated in FIGS. 8A and 8B is a single gate transistor; however, the structure of the transistor is not limited thereto. For example, a multi-gate transistor may be applied.

Next, a process of manufacturing the transistor 410 is described with reference to FIGS. 9A to 9E.

First, the insulating layer 407 serving as a base film is formed over the substrate 400. The insulating layer 407 is preferably formed while moisture remaining in a treatment chamber is removed. This is for preventing hydrogen, water, hydroxyl group, hydroxide, or the like from being contained in the insulating layer 407.

Then, an oxide semiconductor layer is formed over the insulating layer 407 by a sputtering method. Note that before the oxide semiconductor layer is formed, the substrate 400 over which the insulating layer 407 is formed is preferably preheated. This is for preventing hydrogen, moisture, and hydroxyl group from being contained in the oxide semiconductor layer as much as possible. By the preheating, impurities such as hydrogen or moisture adsorbed on the substrate 400 are eliminated and exhausted.

As a target for forming the oxide semiconductor layer, a metal oxide target containing zinc oxide as its main component can be used. For example, a target with a composition ratio of  $\text{In}_2\text{O}_3:\text{Ga}_2\text{O}_3:\text{ZnO}=1:1:1$ , that is,  $\text{In}:\text{Ga}:\text{Zn}=1:1:0.5$  can be used. Alternatively, a target having a composition ratio of  $\text{In}:\text{Ga}:\text{Zn}=1:1:1$  or  $\text{In}:\text{Ga}:\text{Zn}=1:1:2$  can be used.

Alternatively, a target of a metal oxide such as an In—Sn—Ga—Zn—O-based metal oxide, an In—Sn—Zn—O-based metal oxide, an In—Al—Zn—O-based metal oxide, an Sn—Ga—Zn—O-based metal oxide, an Al—Ga—Zn—O-based metal oxide, an Sn—Al—Zn—O-based metal oxide, an In—Zn—O-based metal oxide, an Sn—Zn—O-based metal oxide, an Al—Zn—O-based metal oxide, a Zn—Mg—O-based metal oxide, an Sn—Mg—O-based metal oxide, an In—Mg—O-based metal oxide, an In—O-based metal oxide, an Sn—O-based metal oxide, or a Zn—O-based metal oxide can be used.

Note that as the oxide semiconductor layer, a thin film containing an oxide semiconductor expressed by  $\text{InMO}_3(\text{ZnO})_m$  ( $m>0$ ) can be used. Here,  $M$  represents one or more metal elements selected from Ga, Al, Mn, and Co. For example, Ga, Ga and Al, Ga and Mn, or Ga and Co can be given as  $M$ .

The oxide semiconductor layer is processed in a first photolithography step, so that an island-shaped oxide semiconductor layer 412 is formed (see FIG. 9A). After that, in order to remove hydrogen, water, and hydroxyl group from the oxide semiconductor layer 412, the substrate 400 over which the oxide semiconductor layer 412 is formed is put into an electric furnace and subjected to heat treatment. This heat treatment has an effect of dehydration or dehydrogenation on the oxide semiconductor layer 412.

The temperature of the heat treatment is higher than or equal to  $400^\circ\text{C}$ . and lower than or equal to  $750^\circ\text{C}$ .

preferably higher than or equal to 400° C. and less than the strain point of the substrate. In addition, the atmosphere of the heat treatment does not contain water, hydrogen, or the like.

After the heat treatment, successive heat treatment is preferably performed in an oxygen atmosphere or an atmosphere containing nitrogen and oxygen (e.g., nitrogen to oxygen is 4 to 1 in volume ratio). This is for repairing oxygen deficiency generated in the oxide semiconductor layer 412.

In FIG. 9B, a state where the first electrode 415a and the second electrode 415b are formed over the insulating layer 407 and the oxide semiconductor layer 412 is illustrated. The first electrode 415a functions as one of a source electrode and a drain electrode. The second electrode 415b functions as the other of the source electrode and the drain electrode.

In FIG. 9C, a state where the gate insulating layer 402 is formed over the insulating layer 407, the oxide semiconductor layer 412, the first electrode 415a, and the second electrode 415b is illustrated. Note that it is preferable that hydrogen be not contained in an atmosphere used for forming the gate insulating layer 402.

In FIG. 9D, a state where an opening 421a reaching the first electrode 415a and an opening 421b reaching the second electrode 415b are formed by removal of part of the gate insulating layer 402 is illustrated.

In FIG. 9E, a state where the gate electrode 411, the first wiring layer 414a, and the second wiring layer 414b are formed over the gate insulating layer 402 and in the opening 421a and the opening 421b is illustrated.

In the above manner, a transistor including an intrinsic or substantially intrinsic oxide semiconductor can be manufactured.

This application is based on Japanese Patent Application serial no. 2010-010382 filed with Japan Patent Office on Jan. 20, 2010, the entire contents of which are hereby incorporated by reference.

What is claimed is:

1. An electronic device comprising:
  - a display including a first transistor;
  - a circuit storing a signal; and
  - a rechargeable battery,
 wherein the display and the circuit are driven by electric power stored in the rechargeable battery,
  - wherein the circuit includes a second transistor, a third transistor and a capacitor,
  - wherein one of a source and a drain of the second transistor is electrically connected to a first wiring,

wherein the other of the source and the drain of the second transistor is electrically connected to a second wiring, wherein a gate of the second transistor is electrically connected to one of a source and a drain of the third transistor, and to one electrode of the capacitor, wherein the other of the source and the drain of the third transistor is electrically connected to a third wiring, wherein a gate of the third transistor is electrically connected to a fourth transistor, wherein the other electrode of the capacitor is electrically connected to a fifth wiring, wherein a channel of the first transistor contains a first oxide semiconductor, wherein a channel of the second transistor contains silicon, and wherein a channel of the third transistor contains a second oxide semiconductor.

2. The electronic device according to claim 1, wherein the electronic device is an e-book reader.

3. An electronic device comprising:
 

- a display including a first transistor;
- a circuit storing a signal; and
- a rechargeable battery capable of contactless charge,

 wherein the display and the circuit are driven by electric power stored in the rechargeable battery, wherein the circuit includes a second transistor, a third transistor and a capacitor, wherein one of a source and a drain of the second transistor is electrically connected to a first wiring, wherein the other of the source and the drain of the second transistor is electrically connected to a second wiring, wherein a gate of the second transistor is electrically connected to one of a source and a drain of the third transistor, and to one electrode of the capacitor, wherein the other of the source and the drain of the third transistor is electrically connected to a third wiring, wherein a gate of the third transistor is electrically connected to a fourth transistor, wherein the other electrode of the capacitor is electrically connected to a fifth wiring, wherein a channel of the first transistor contains a first oxide semiconductor, wherein a channel of the second transistor contains silicon, and wherein a channel of the third transistor contains a second oxide semiconductor.

4. The electronic device according to claim 3, wherein the electronic device is an e-book reader.

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